



Political *Science*

Why Is It “Political *Science*” Rather Than Just “Politics?”

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POLS 222: Slide Deck #3

Where We Left Off

Humans are good at thinking causally and probabilistically.

Ordinary Human Inquiry:

- prediction without understanding (tradition and authority)
- relies on individual experiences, second-hand knowledge
- prone to errors
 - inaccurate observations, overgeneralization, selective observation, illogical reasoning, ideology

Scientific Inquiry:

- prediction *and* understanding
- *objective* logic and direct *empirical* observation help minimize errors
- examines relationships between variables systematically
- seeks patterns to help with description, prediction, and prescription

Asking the Right Questions

A good research question:

- uses a little bit of background knowledge
- begins with an empirical puzzle (“hmmm...”)
- uses “reporter” questions (e.g. why, what, how)
- links to or draws from theories in the field
- are about outcomes rather than causes, usually

A less good research question:

- have a one-sentence or factual answer
- contains a lot of proper nouns
- has a single correct answer
- focuses heavily on specific measures/indicators

We're in the business of explaining *and* predicting!

- We need to be able to generalize
- Variables provide concept abstraction to help find patterns
- Focusing on a case can lead to errors of inquiry
- What if the case is an outlier?

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- A specific voter named Voter Votingham identifies as moderate
- Student Studentson is present today and earning a B in class
- The Republic of Equatorial Kundu is an autocracy

Different Functions of Different Variables

The basics:

- **Dependent variable (Y):** The phenomenon we want to predict
 - Think of it like the outcome or maybe "the effect"
 - There is almost always only one DV in a model.
- **Independent variable (X→Y):** The phenomenon we think explains the dependent variable
 - Think of it like the input or maybe "the cause" of that effect.
 - There can be several IVs in a model. (e.g. $X_1, X_2, X_3 \rightarrow Y$)

Some more complicated types:

- Intervening variable: a hypothetical phenomenon that comes between the DV and an IV ($X \rightarrow Z \rightarrow Y$)
- Confounding variable: a phenomenon that influences both the DV and the IV ($Z \rightarrow X$, and $Z \rightarrow Y$)
- Control variable: a phenomenon that influences the relationship between IV and DV whose effects are canceled out.

Let's Practice...

What are some concepts we want to explain?

What are some concepts we want to explain it with?

Explaining and predicting requires hypothesis testing.

Part of evaluation is re-testing, generating additional hypotheses, or reworking original hypotheses.

Skepticism and transparency are key

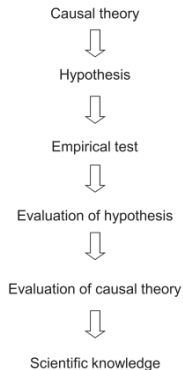


Figure 1.1 The road to scientific knowledge

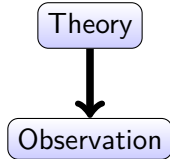
Rules of the Road

1. Focus on Causality
2. Don't Let Data Alone Drive Your Theories
3. Consider Only Empirical Evidence
4. Check your Ideology at the Door
5. Pursue Both Generality and Parsimony

Deduction vs. Induction

Deductive reasoning:

- How well do cases support the theory?



Inductive reasoning:

- What is this a case of?

